

JALEN EDUSEI

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EDUCATION

University of Georgia, Morehead Honors College

Athens, GA

B.S. Computer Systems Engineering | GPA: 3.61

Aug. 2022 – May 2026

- **Relevant Coursework:** Data Structures, Systems Programming, Software Development, Discrete Mathematics, Advanced Digital Design, Embedded Systems I & II, Design of Digital Systems, Sensors & Transducers, Linear Systems, Human-Computer Interaction.

EXPERIENCE

Business Analyst Intern

June 2025 – Aug. 2025

Capital One

McLean, VA

- Built a data-driven business case for a Notifications Preferences Center on the CreditWise team, projected to streamline communication management for 60M+ users.
- Leveraged SQL, Python, and Excel to analyze email-campaign KPIs and engineer a valuation framework quantifying engagement and retention impact, informing product roadmap decisions.
- Partnered with cross-functional product, design, and engineering teams to present recommendations to senior leadership and align on future messaging strategy.

Flight Software Developer — MEMESat-1 CubeSat

Mar. 2024 – Dec. 2024

UGA Small Satellite Research Laboratory

Athens, GA

- Developed and debugged embedded flight software in C/C++ for the MEMESat-1 CubeSat on STM32 microcontrollers, bringing up and troubleshooting UART communication between onboard subsystems.
- Authored unit and integration tests that hardened the flight system against edge cases and failure conditions, and reviewed teammate code to improve quality across the codebase.

PROJECTS

Live Election Platform | *Next.js, TypeScript, Supabase, PostgreSQL, WebSocket*

Mar. 2026 – Apr. 2026

- Shipped a presenter-paced realtime voting system for 60–80 concurrent users across 14 roles, using a finite-state election machine and Supabase WebSocket broadcast for sub-100ms ballot delivery.
- Engineered a defense-in-depth anti-fraud stack with composite SHA-256 browser fingerprinting, a database UNIQUE constraint, and Postgres Row-Level Security to prevent duplicate voting and tampering.

AnimalDot Smart Bed | *C++, ESP32, BLE/WiFi/MQTT, DSP, React Native*

Aug. 2025 – May 2026

- Engineered ESP32 firmware in C++ for a contactless vital-signs bed, orchestrating geophone, temperature, and load-cell sensors through a modular sensor manager with NVS-persisted calibration.
- Built a real-time DSP pipeline (band-pass filtering, peak detection, smoothing) and concurrent BLE/WiFi/MQTT stacks streaming telemetry to a React Native app, Node.js backend, and Next.js dashboard.

PrimeForge FPGA Prime Detector | *Verilog, Xilinx FPGA, VGA, DDR2, SPI*

Jan. 2026 – May 2026

- Architected a Verilog prime-detection system unifying dual compute cores (Sieve + trial division), a 640x480 VGA pipeline, SD-card SPI interface, and DDR2 arbiter, verified with module-level testbenches.

Immersive 3D Portfolio | *React Three Fiber, TypeScript, Zustand, Tailwind CSS*

Jan. 2026 – Present

- Built a 3D “workstation” portfolio with a custom theme engine and terminal-inspired navigation, optimized to a consistent 60 FPS PWA across browsers and deployed end-to-end on Vercel.

TECHNICAL SKILLS

Languages: C, C++, C#, Python, TypeScript, JavaScript, Java, SQL, Verilog, Assembly, MATLAB

Frameworks & Tools: React, Next.js, Node.js, React Native, React Three Fiber, Tailwind CSS, Supabase, PostgreSQL, Unity, Git, Figma, Vercel

Embedded & Hardware: STM32, ESP32, Raspberry Pi, FPGA (Xilinx), BLE/NimBLE, MQTT, UART/SPI, RTOS concepts, signal processing, NASA F' (F Prime)

AI: Anthropic Claude API, LLM application development, tool use, structured (JSON) outputs, prompt engineering

LEADERSHIP & HONORS

Vice President, National Society of Black Engineers (NSBE) — UGA Chapter (100+ members) | Tau Beta Pi Honor Society | Resident Assistant | *Extraordinary Engineer* & 2x Presidential Scholar awards.